

Question:

Jam has three boxes full of pencils and 2 loose pencils which give a total of 26 pencils. If her sister, Meg, has 46 pencils, how many boxes do Jam and Meg need to store all their pencils?

Reasoning + Answer:

<think>First let's determine how many pencils Jam has in total . Jam has 26 pencils in total . Out of which 2 pencils are loose . Therefore Jam has $26 - 2 = 24$ pencils in the boxes . Now let's determine how many boxes Jam has . We know that Jam has three boxes full of pencils . Therefore Jam has 3 boxes in total . Now let's calculate the total number of pencils Jam and Meg have . Jam has 26 pencils . Meg has 46 pencils . Therefore in total they have $26 + 46 = 72$ pencils . Now let's calculate how many boxes they need to store all their pencils . We know that Jam has 3 boxes . Meg has 46 pencils . From the previous calculation we know that there are 24 pencils in Jam's boxes . Therefore Meg needs $46 - 24 = 22$ pencils in boxes . Since each box contains the same number of pencils as Jam's boxes . Each box can store $24 / 3 = 8$ pencils . Now let's calculate how many boxes are needed to store 22 pencils . $22 / 8 = 2.75$. Since we can't have a fraction of a box . round up to the nearest whole number . Therefore Meg needs 3 boxes to store her 22 pencils . In total Jam has 3 boxes . Meg needs 3 boxes . Therefore in total they need $3 + 3 = 6$ boxes to store all their pencils . </think><answer>6</answer>

